

Name: Solutions.

Teacher: _____

Class: _____



Baldivis
Secondary College

**MATHEMATICS – PATHWAY 1
YEAR 10 EXAM
SEMESTER TWO 2020**

Section Two: Resource Allowed

TIME ALLOWED FOR THIS PAPER

Reading time before commencing 5 minutes.

Working time for paper 60 minutes.

MATERIAL(S) REQUIRED / RECOMMENDED FOR THIS PAPER

TO BE PROVIDED BY THE SUPERVISOR:

This Question / Answer Booklet
Formula Sheet

TO BE PROVIDED BY THE CANDIDATE:

Standard items: pens, pencils, pencil sharpener, eraser, correction fluid, ruler, highlighters.

Special items: Calculator, 1 x A4 sheet of notes.

Important note to candidates:

No other items may be used in this section of the examination. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Instructions to candidates

- Candidates are to remain seated during the exam and are not allowed to talk to other candidates.
- Mobile phones are not to be used at all during the exam.
- Students who fail to follow examination rules will be removed and dealt with in accordance to the schools policy.

Write your answers in the spaces provided in this Question/Answer Booklet. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use spare pages for planning, indicate this clearly at the top of the page
- Continuing an answer: If you need the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Show all your working clearly. Your working should be sufficient to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat an answer to any question, ensure that you cancel the answer you do not wish to have marked.

Use a blue or black pen only, except with diagrams where you should use pencil.

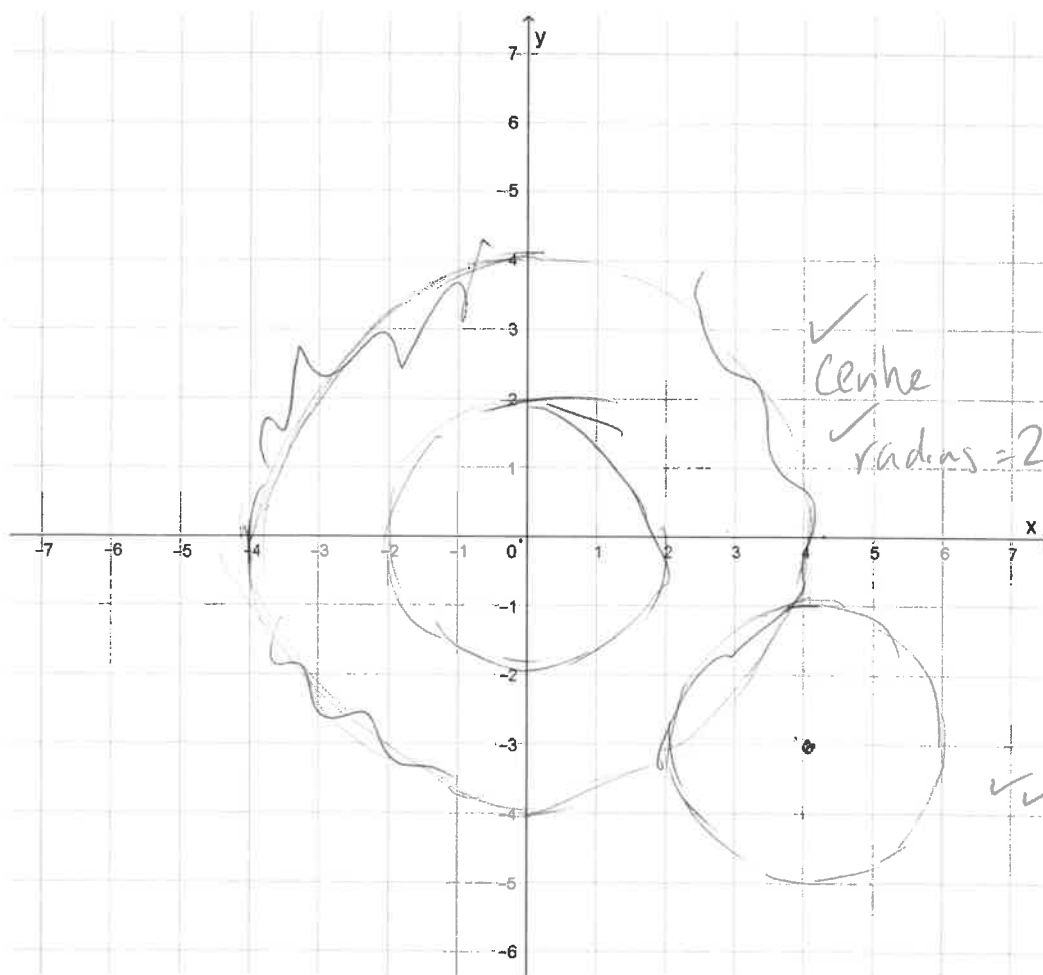
Question	Out of	Mark
5	6	
6	6	
7	9	
8	9	
9	6	
10	7	
11	8	
12	3	
13	7	
TOTAL	61	
	Mark	
Percentage		

Question 5

(6 marks)

- On the grid below sketch the circle $x^2 + y^2 = 4$.
- This circle is translated 4 units right and 3 units down. Sketch this circle on the grid.
- What is the equation of this circle?

$$(x-4)^2 + (y+3)^2 = 4$$



✓

Question 6

(6 marks)

Solve the following simultaneous equations

a) $3x + 2y = 19$
 $y = 2x - 8$

$$3x + 2(2x - 8) = 19$$

$$3x + 4x - 16 = 19$$

$$7x = 35$$

$$x = 5$$

$$y = 2 \times 5 - 8$$

$$y = 2$$

b) $2x + 5y = -1$
 $4x + 3y = 5$

$$4x + 10y = -2$$

$$-4x + 3y = 5$$

$$7y = -7$$

$$y = -1$$

$$2x - 5 = -1$$

$$2x = 4$$

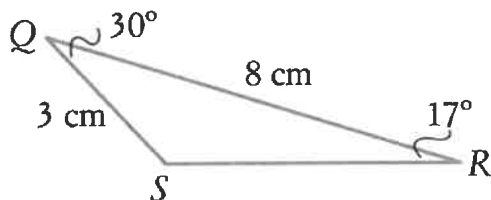
$$x = 2$$

Question 7

(9 marks)

- a) Find the area of this triangle.

(2 marks)



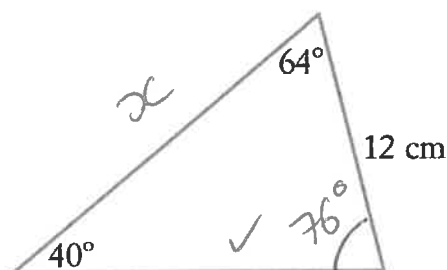
$$\text{Area} = \frac{1}{2} \times 3 \times 8 \times \sin 30^\circ$$

$$= \frac{1}{2} \times 3 \times 8 \times \frac{1}{2}$$

$$= 6 \text{ cm}^2$$

- b) Find the length of the longest side of this triangle.

(4 marks)



$$\frac{x}{\sin 76^\circ} = \frac{12}{\sin 40^\circ}$$

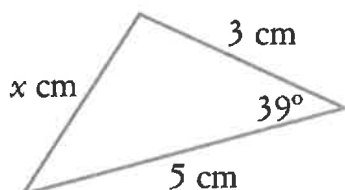
Sine rule ✓

$$x = \frac{12 \sin 76^\circ}{\sin 40^\circ}$$

$$x = 18.1 \text{ cm}$$

- c) Find the length marked x cm.

(3 marks)



$$x^2 = 3^2 + 5^2 - 2 \times 3 \times 5 \cos 39^\circ$$

$$x^2 = 10.69 \text{ cm}^2$$

$$x = 3.27$$

Cosine rule ✓

Question 8

(9 marks)

Consider the equation $y = x^2 - 2x - 8$

- a) Find the coordinate of the y-intercept.

$(0, -8)$ ✓

(1 mark)

- b) Find the coordinates of the x-intercepts.

$$(x-4)(x+2) = 0$$

$$x=4 \quad x=-2$$

$(4, 0), (-2, 0)$ (3 marks)

- c) What are the coordinates of the minimum point?

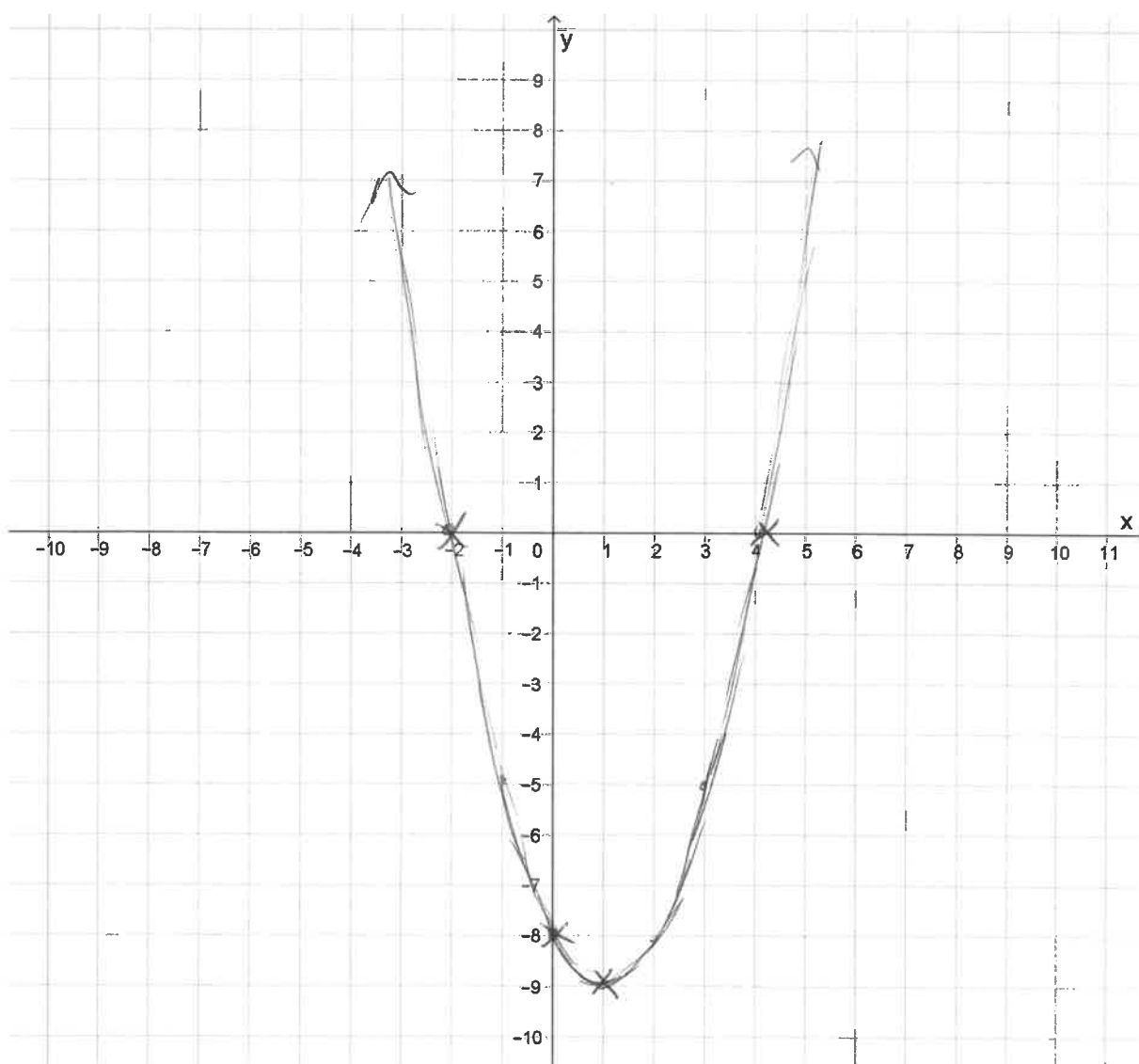
$$y = (x-1)^2 - 9$$

$$(1, -9)$$

(3 marks)

- d) Sketch the graph of the grid below

(2 marks)



- 1/2 mark for each point not plotted
✓ curve joining points.

Question 9

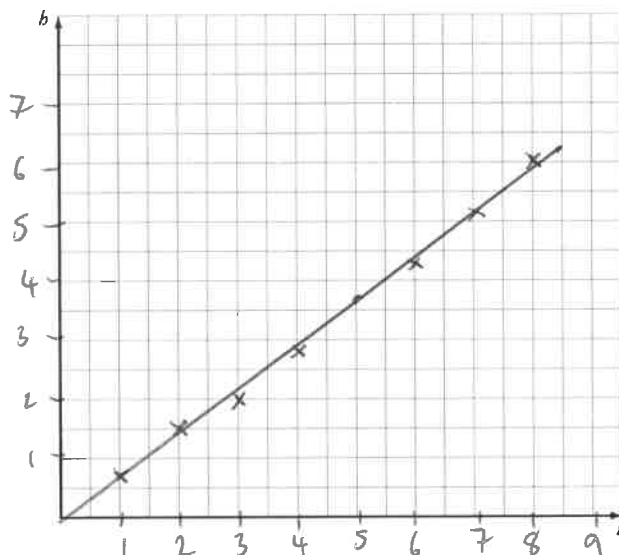
(6 marks)

Michael measured the height of a seedling after the first shoot appeared and recorded his results in the table below.

Day	1	2	3	4	5	6	7	8
Height (cm)	0.7	1.5	2.0	2.8		4.3	5.1	6.0

a) Plot the points on the grid below.

(3 marks)



axes labelled ✓
points ✓✓
-1/2 each error.

b) Describe the relationship between the day and the height.

(1 mark)

The height increases each day. ✓

c) Draw in the line of best fit.

(1 mark)

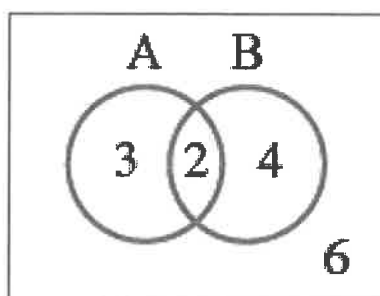
d) Michael made a mistake on day 5 and did not include this in the table. Estimate the height of the seedling on day 5.

(1 mark)

≈ 3.6 ✓

Question 10**(7 marks)**

Consider the following Venn diagram displaying the number of elements belonging to the events A and B.



a) Find:

i. $n(A')$ 10

ii. $n(A' \cap B')$ 6

iii. $n(A \cup B)$ 9

b) Find:

i. $\Pr(A)$ $\frac{5}{15} = \frac{1}{3}$

ii. $\Pr(A \cap B)$ $\frac{2}{15}$

iii. $\Pr(A \cap B)$ $\frac{3}{15} = \frac{1}{5}$

iv. $\Pr(A | B)$

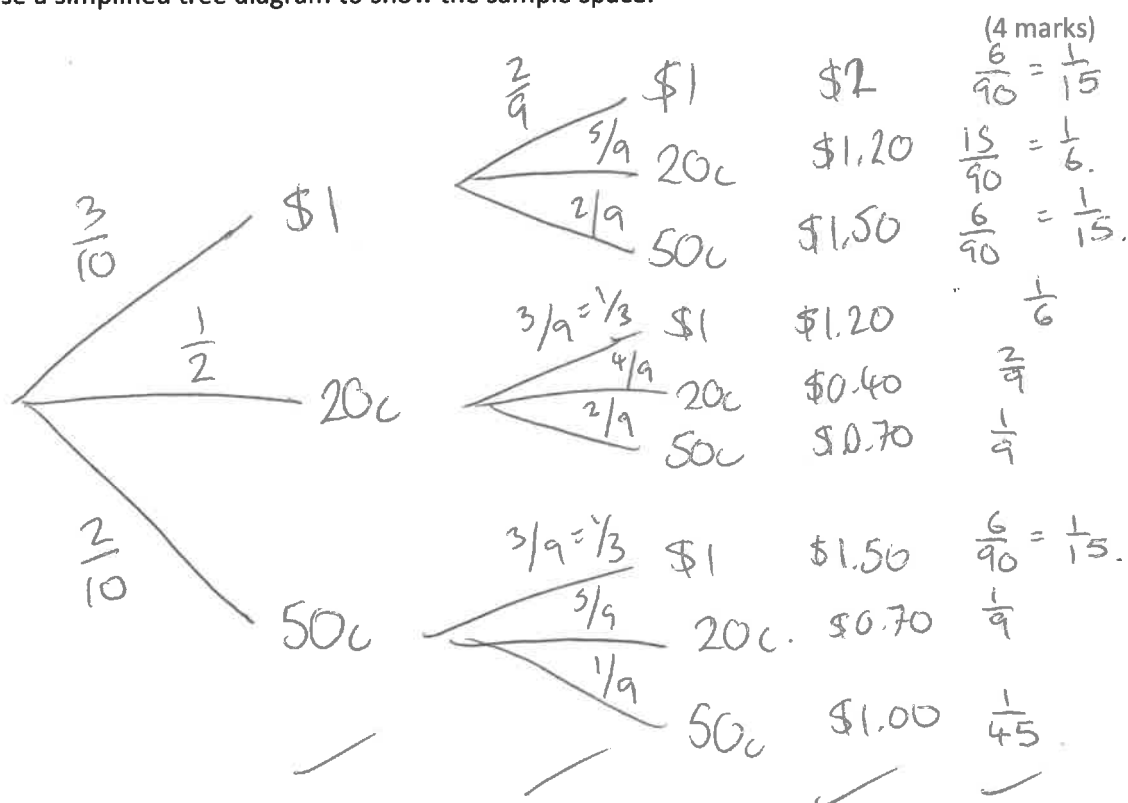
$$\frac{2}{6} = \frac{1}{3}$$

Question 11

(8 marks)

Peta's purse contains three \$1 coins, five 20c coins and two 50c coins. Two coins randomly fall out.

a) Use a simplified tree diagram to show the sample space.



b) Find the probability the two coins will be the same.

(2 marks)

$$\frac{2}{90} + \frac{20}{90} + \frac{6}{90} = \frac{28}{90} = \frac{14}{45}$$

c) Find the probability the exact money to pay for a \$1.20 newspaper will fall out.

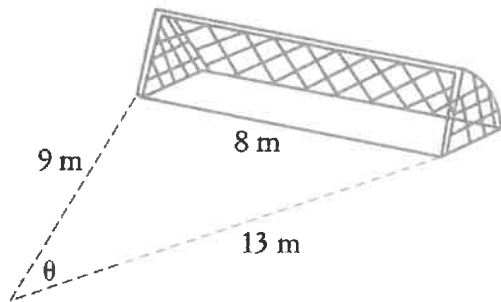
(2 marks)

$$\frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

Question 12

(3 marks)

A soccer player is 9 m from one goal post and 13 m from the other. Within what angle θ must he kick to score? Answer to the nearest degree.



$$\cos \theta = \frac{9^2 + 13^2 - 8^2}{2 \times 9 \times 13}$$

$$\theta = 37^\circ$$

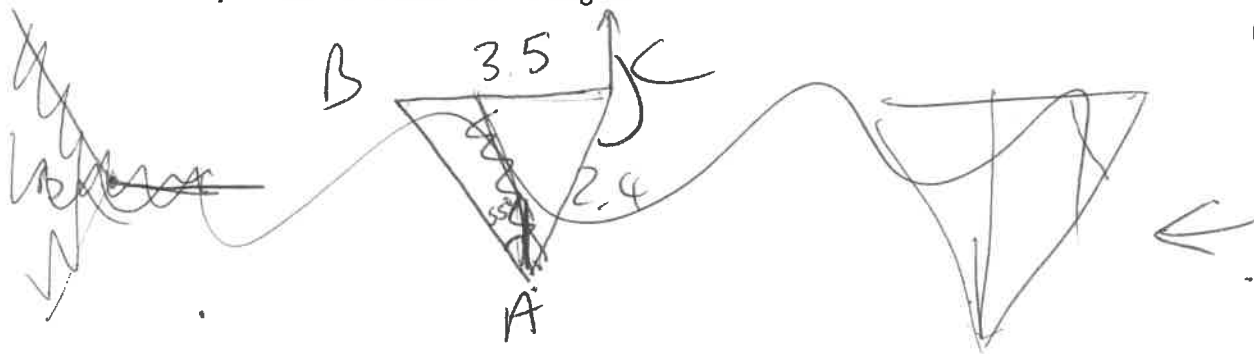
Question 13

(7 marks)

On an orienteering course, Sonya runs from point A to point B at a bearing of 305° . She then runs 3.5 km due east to point C, then 2.4 km directly back to point A. It is required to find the bearing of C from A and the total distance she has run.

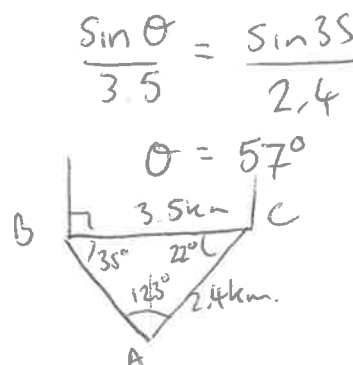
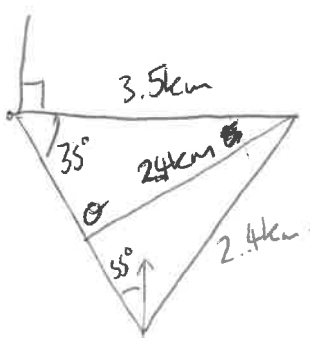
a) Show this situation on a diagram.

(2 marks)



b) Solve the problem and show on a diagram the two possible solutions.

(5 marks)



END OF SECTION 2

C from A
 068° or 002°

	Distance	Distance
AB =	1.55 m	4.18
Total	7.45 m	10.08
	<u>7.5 m</u>	<u>10.1 m</u>

